WEEK-10--Elasticbeanstalk

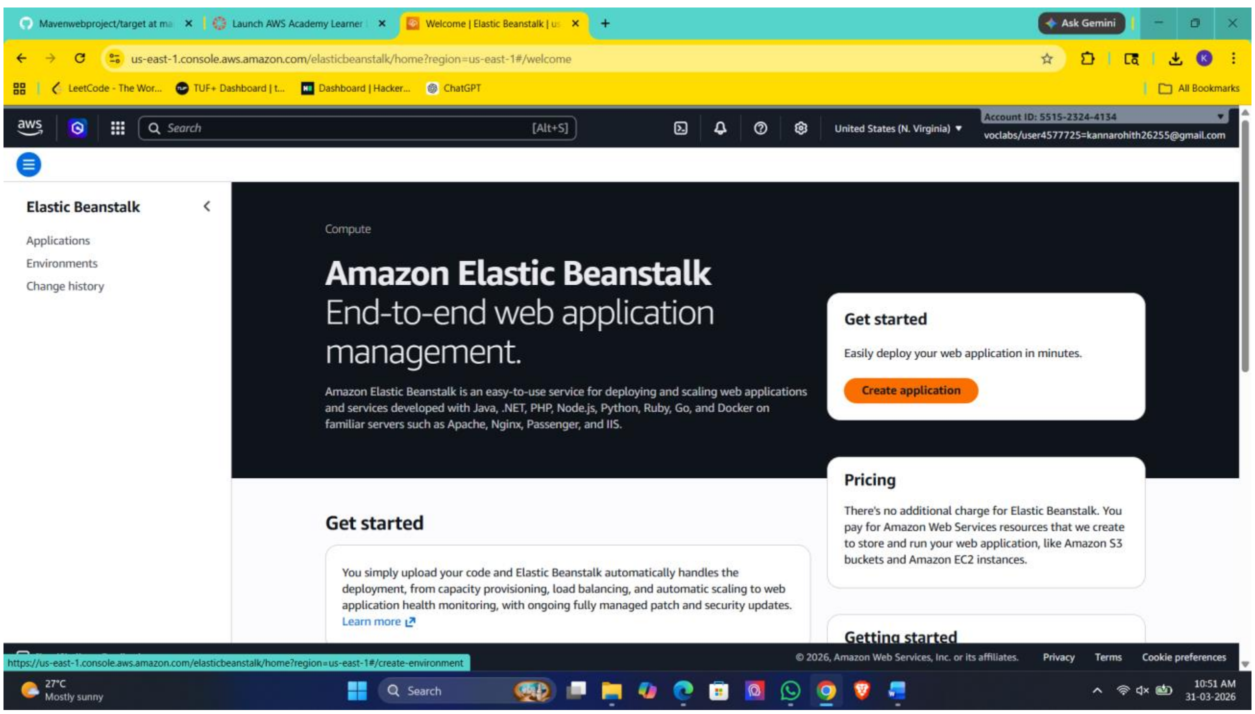
Prerequisites

- A web application ready (e.g., ROOT.war, .zip, or code)----**YOU CAN YOURS MAVEN PROJECT HERE**

Steps to Create Elastic Beanstalk Application

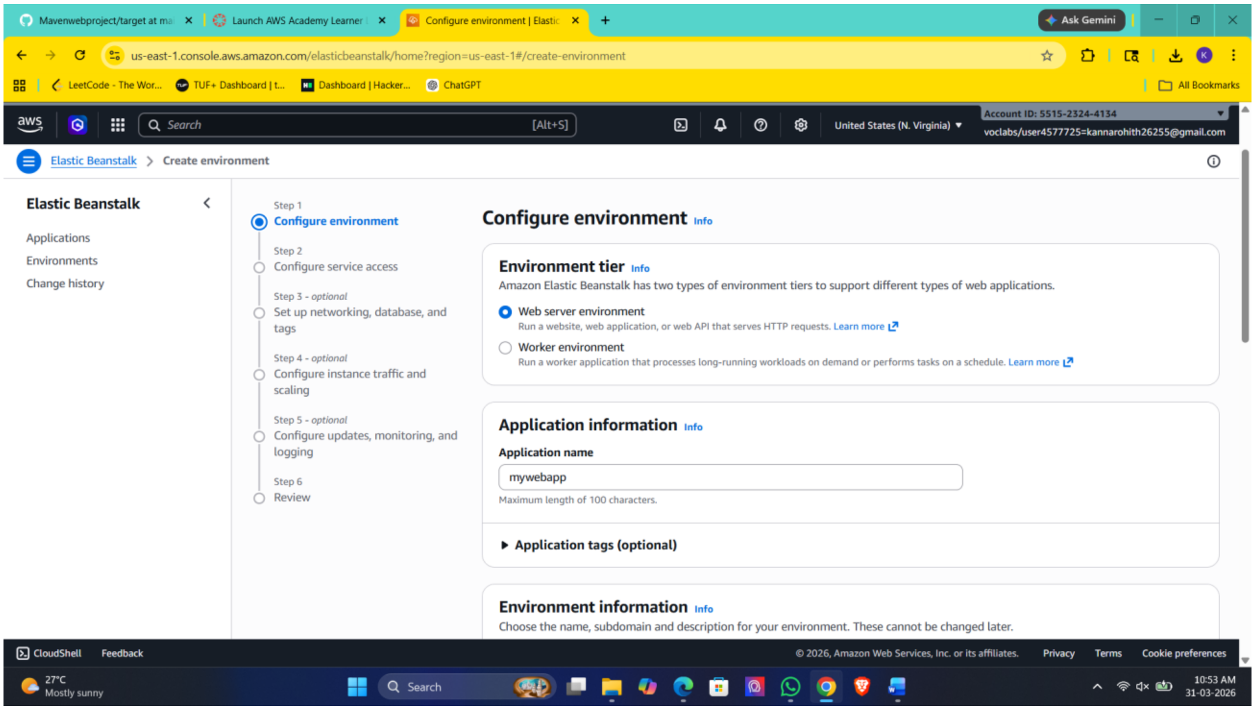
Step 1: Open Elastic Beanstalk

1. Login to AWS Console
2. Under Compute section ->Search for **Elastic Beanstalk**
3. Click **Create Application**



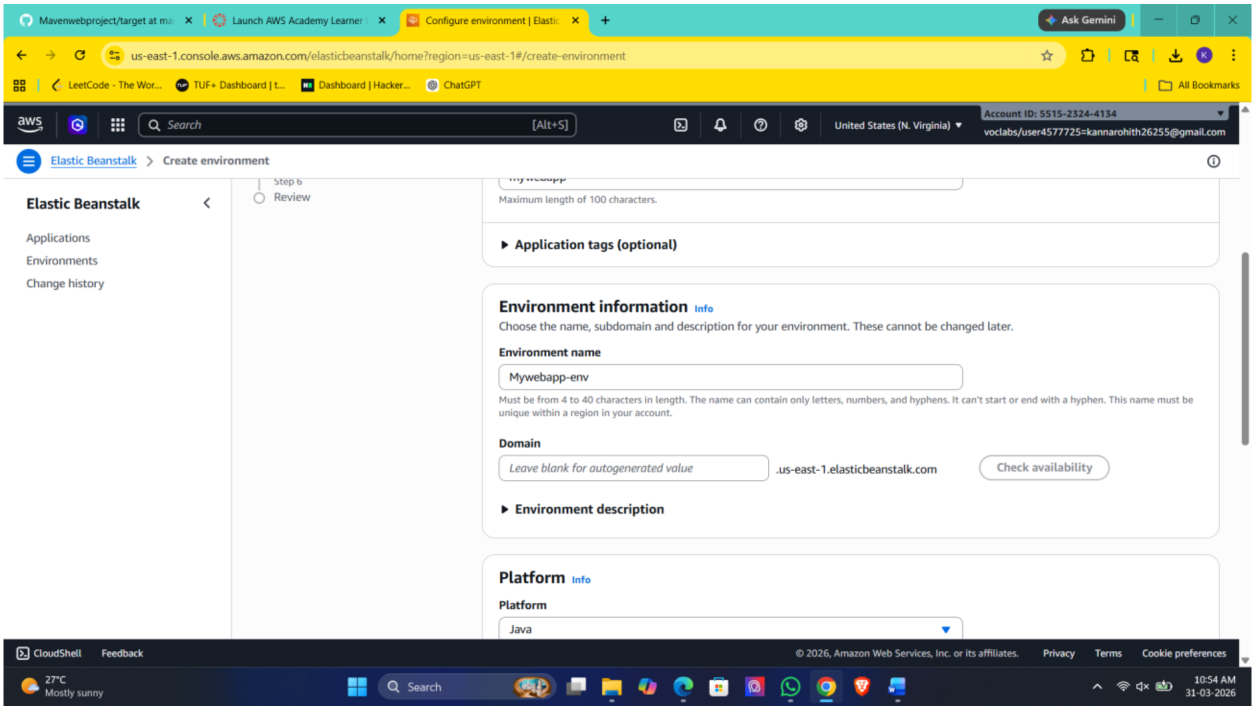
Step 2: Create Application

1. Enter **Application Name** (e.g., MyApp)
2. Add description (optional)
3. Click **Create**



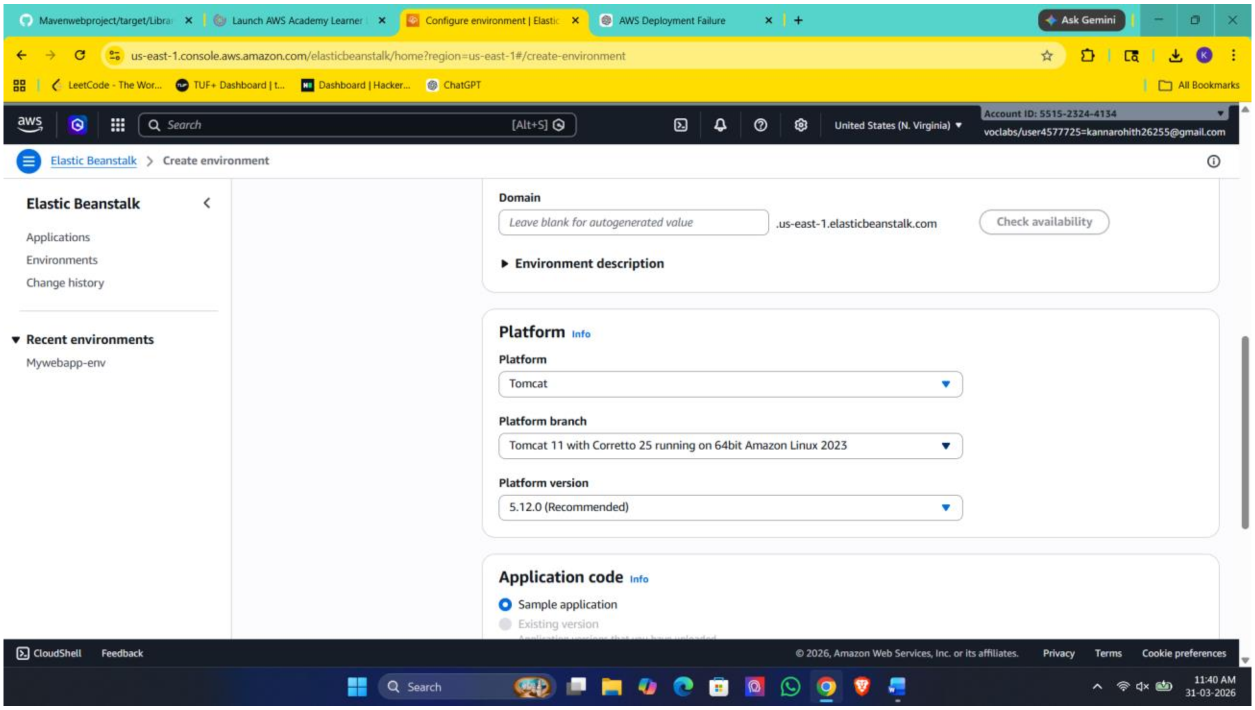
Step 3: Create Environment

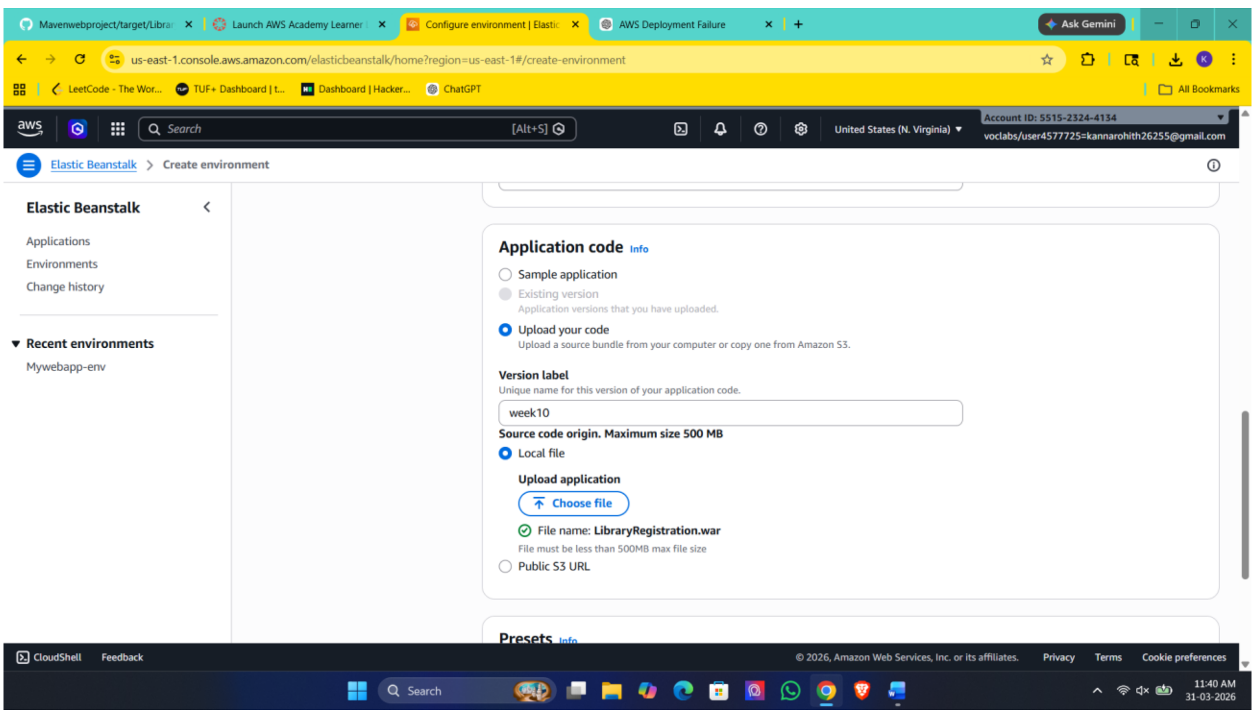
1. Click **Create Environment**
2. Choose:
   - **Web Server Environment** (for web apps)
3. Click **Select**



Step 4: Configure Environment

1. Enter **Environment Name**
2. Choose **Platform**:
   - Java (for .war)
   - Node.js / Python / PHP / etc.
3. Under **Application Code**:
   - Select **Upload your code**
   - Upload your .war or .zip file

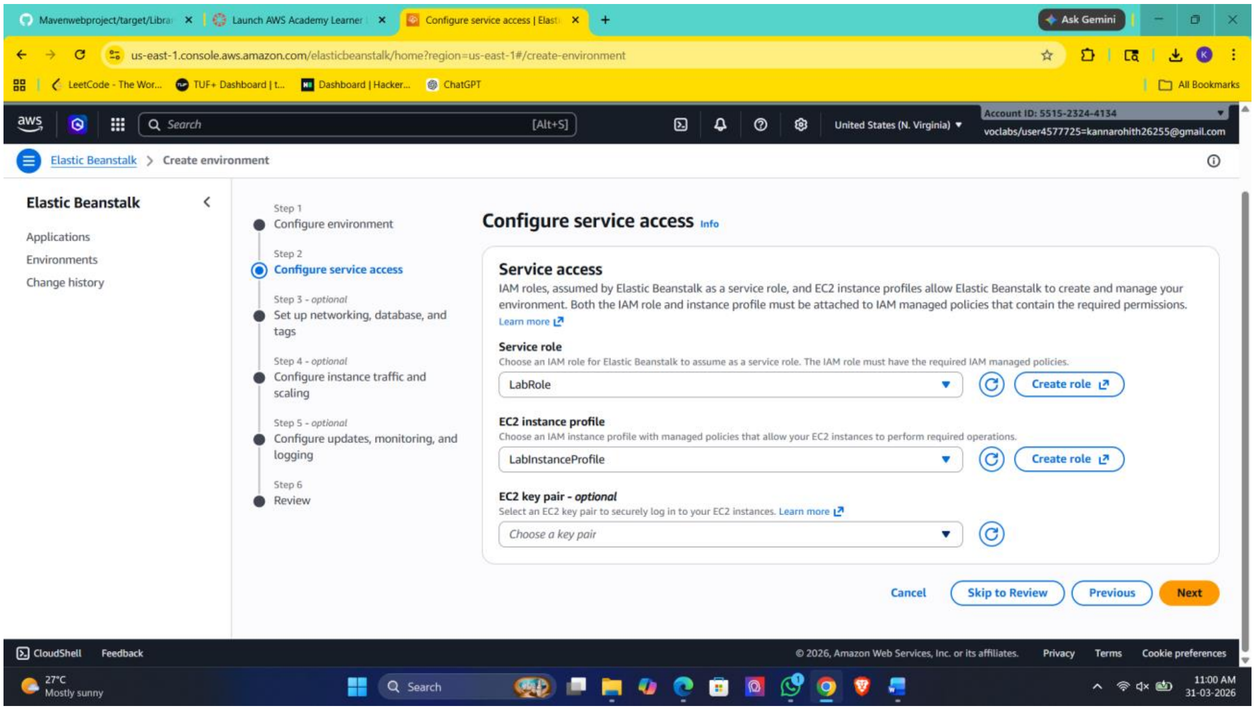




Step 5: Configure Service Access

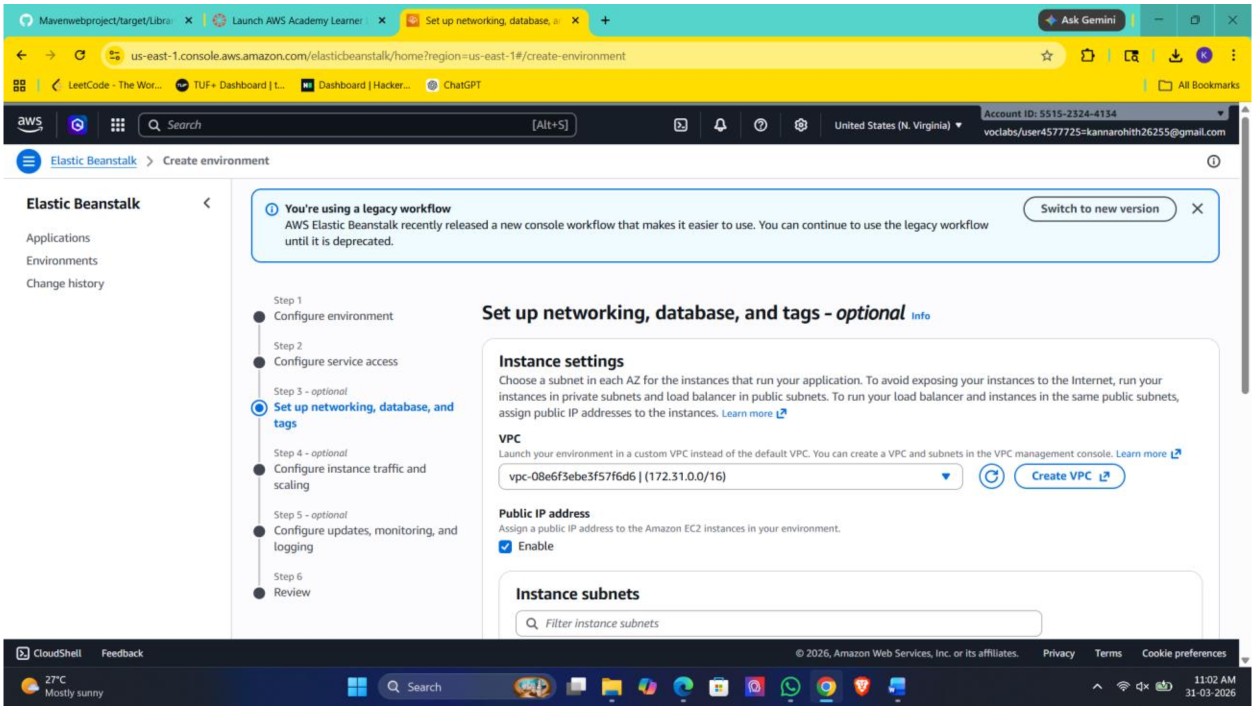
1. Service Role → Select existing -→**LabRole**
2. EC2 Key Pair → Optional (for SSH)

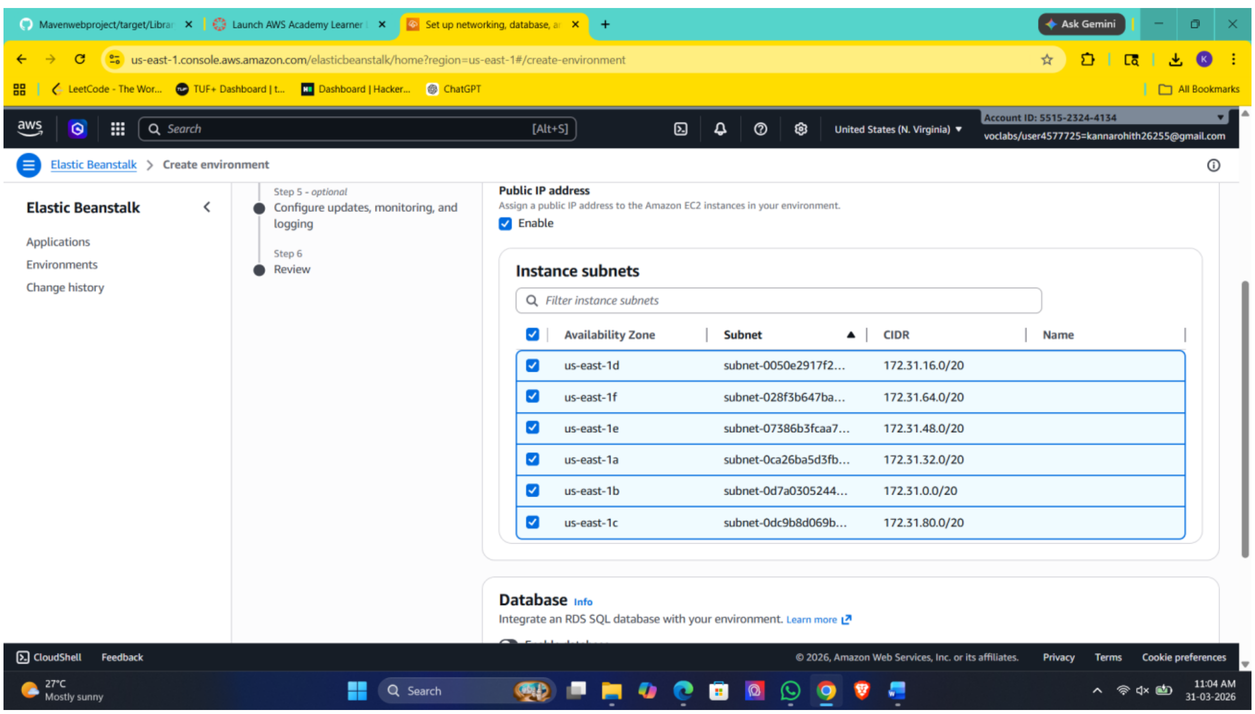
3. EC2 Instance Profile → select (e.g., LabInstanceProfile)



Step 6: Configure Network

1. Select **VPC** (default is fine)
2. Select **Subnets**
3. Enable **Public IP**





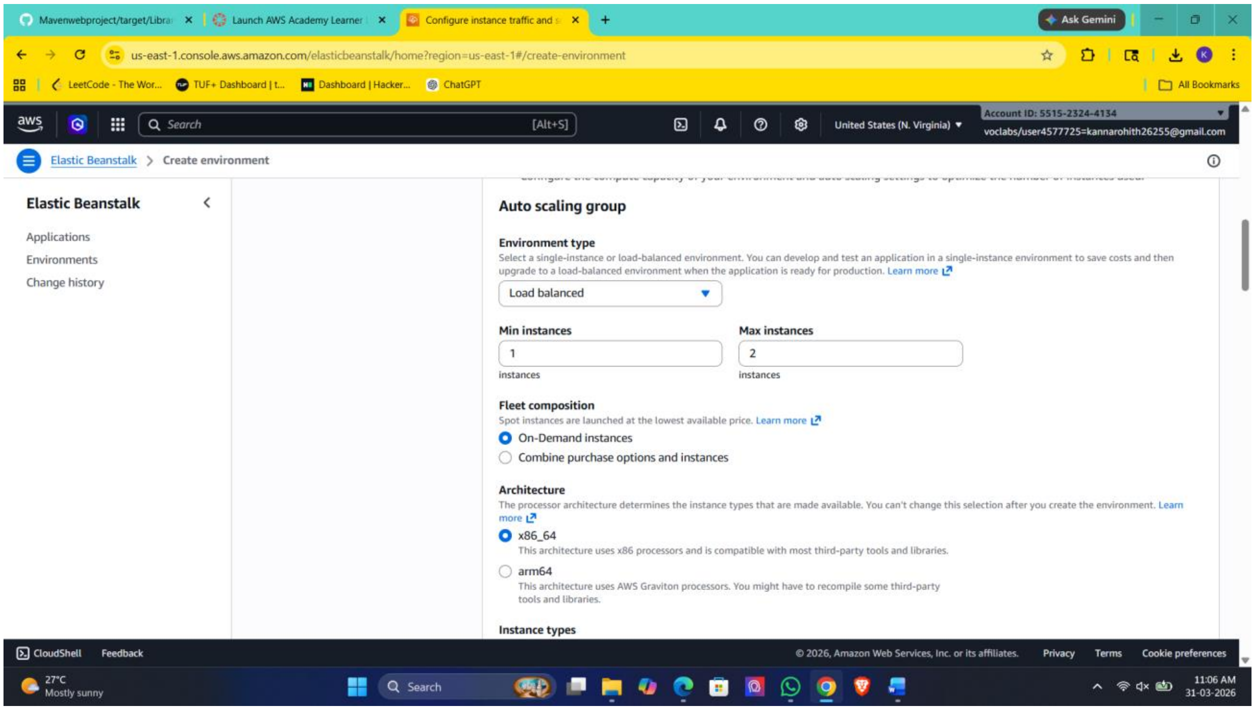
Step 7: Instance & Scaling Setup

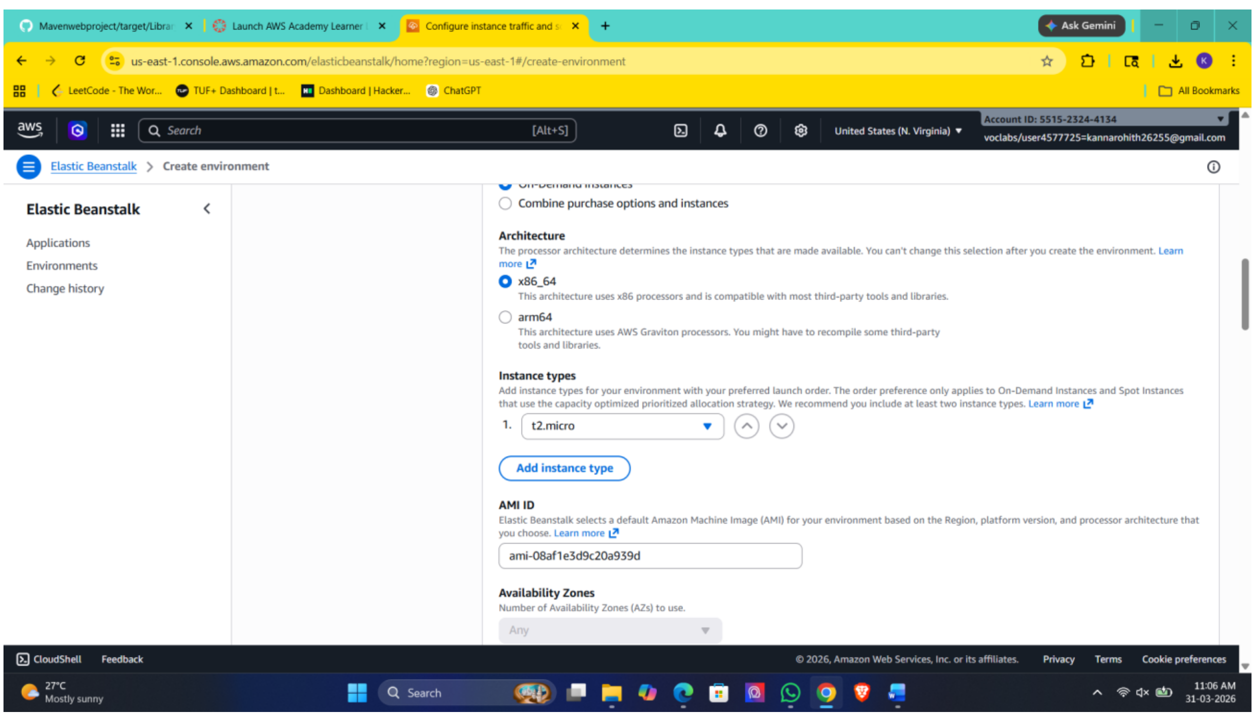
1. Choose **Instance type** (e.g., t2.micro – free tier)

2. Set:

- Min instances: 1
- Max instances: 2 (or more)

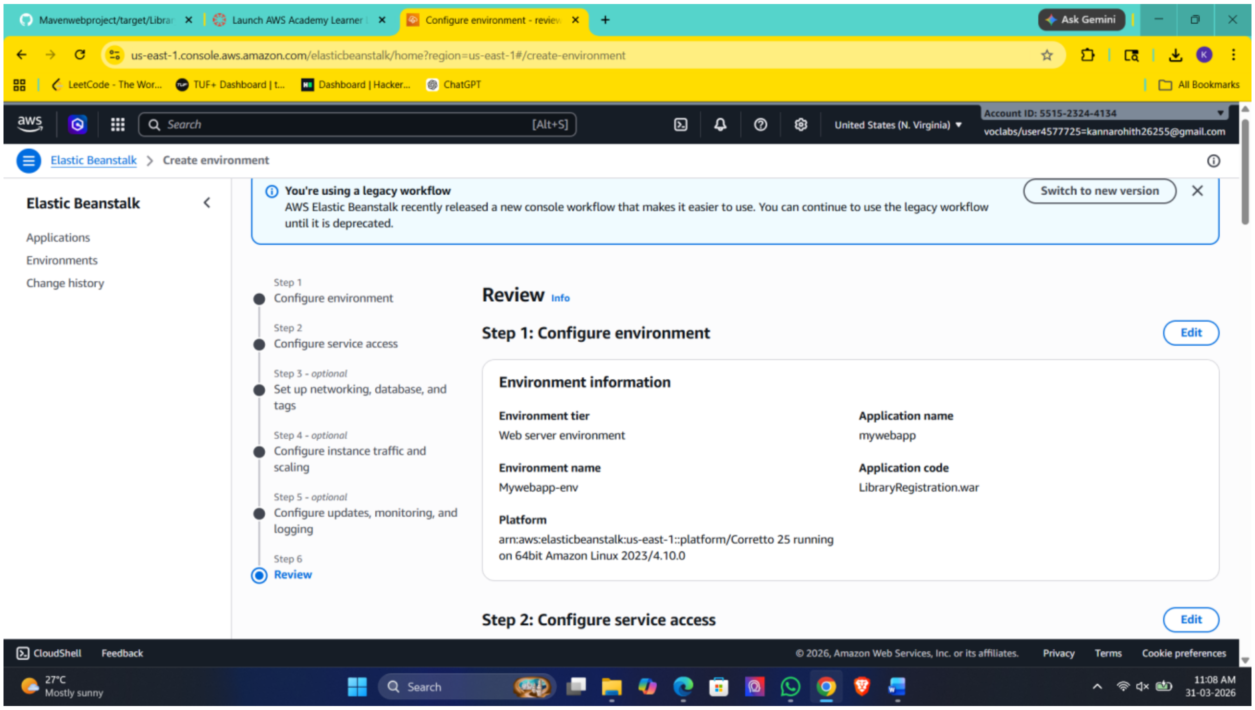
3. Load Balancer → Enabled (optional but recommended)

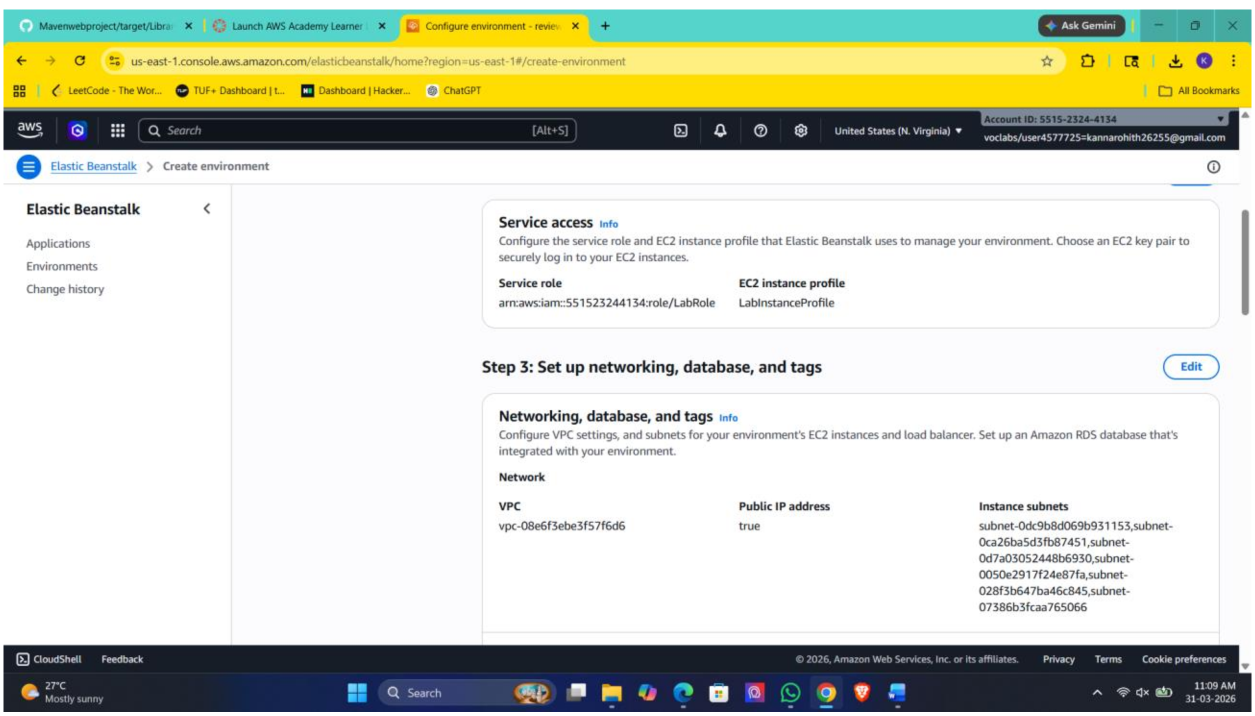


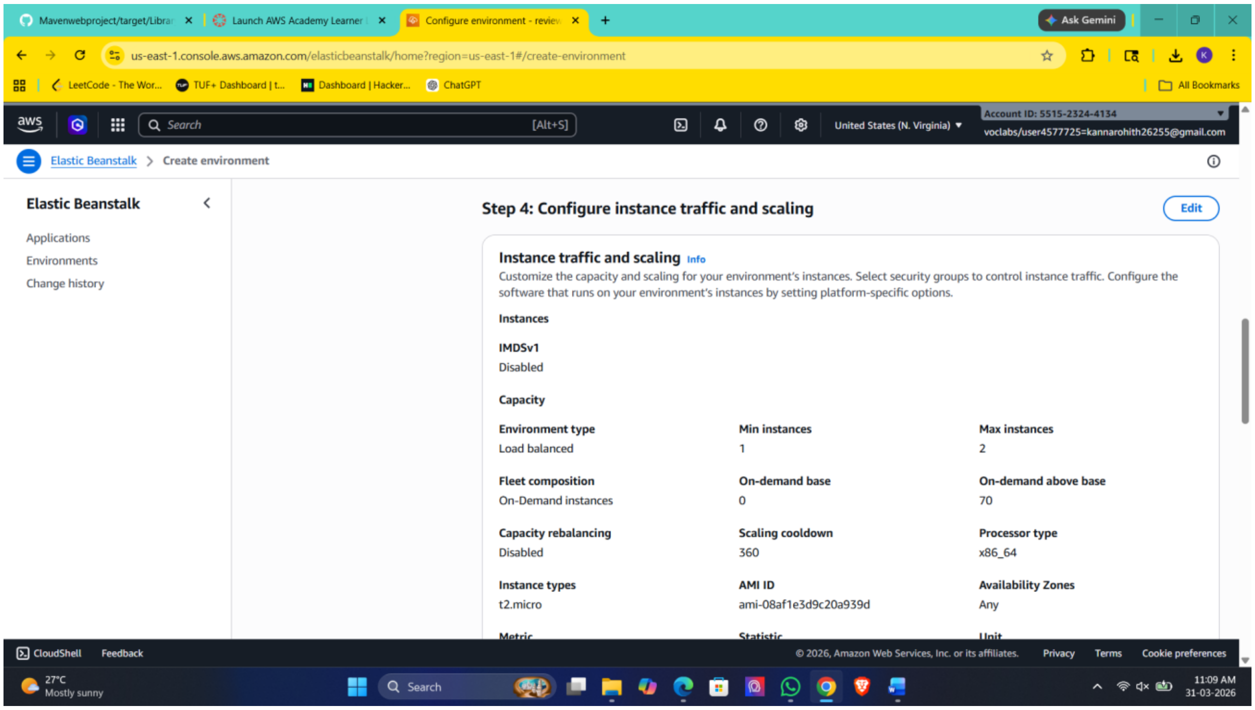


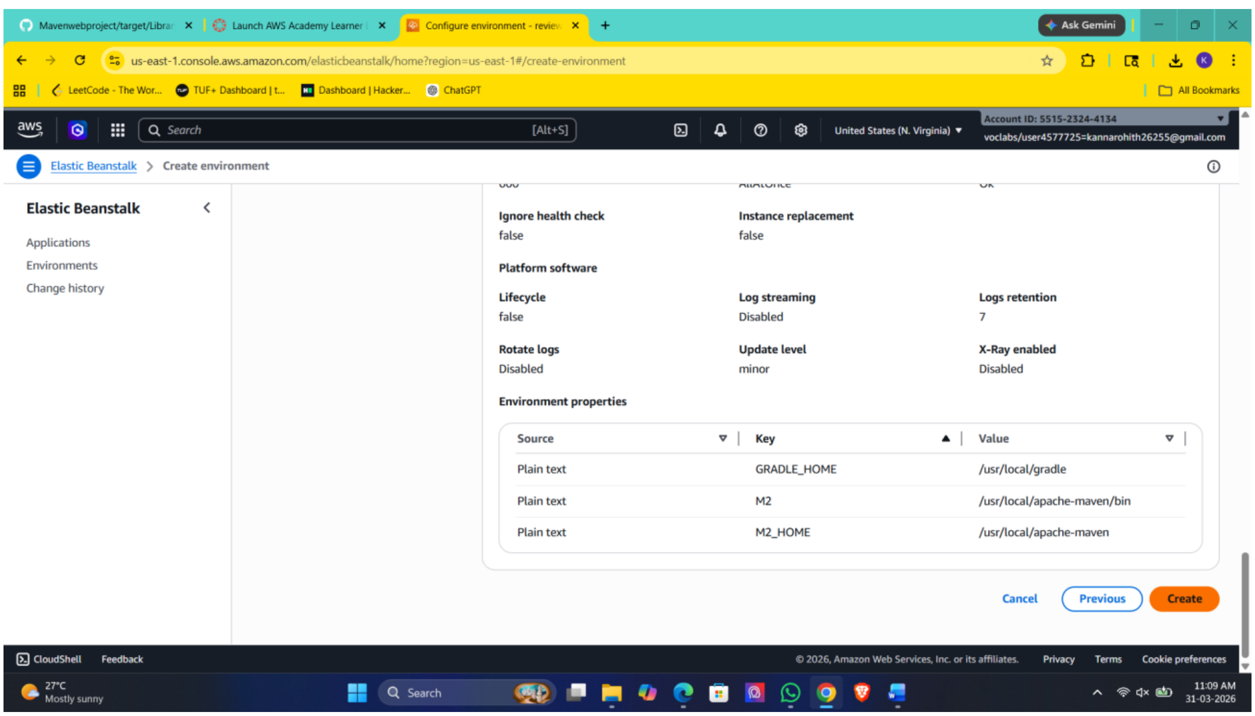
Step 8: Review & Create

1. Click **Review**
2. Click **Create**









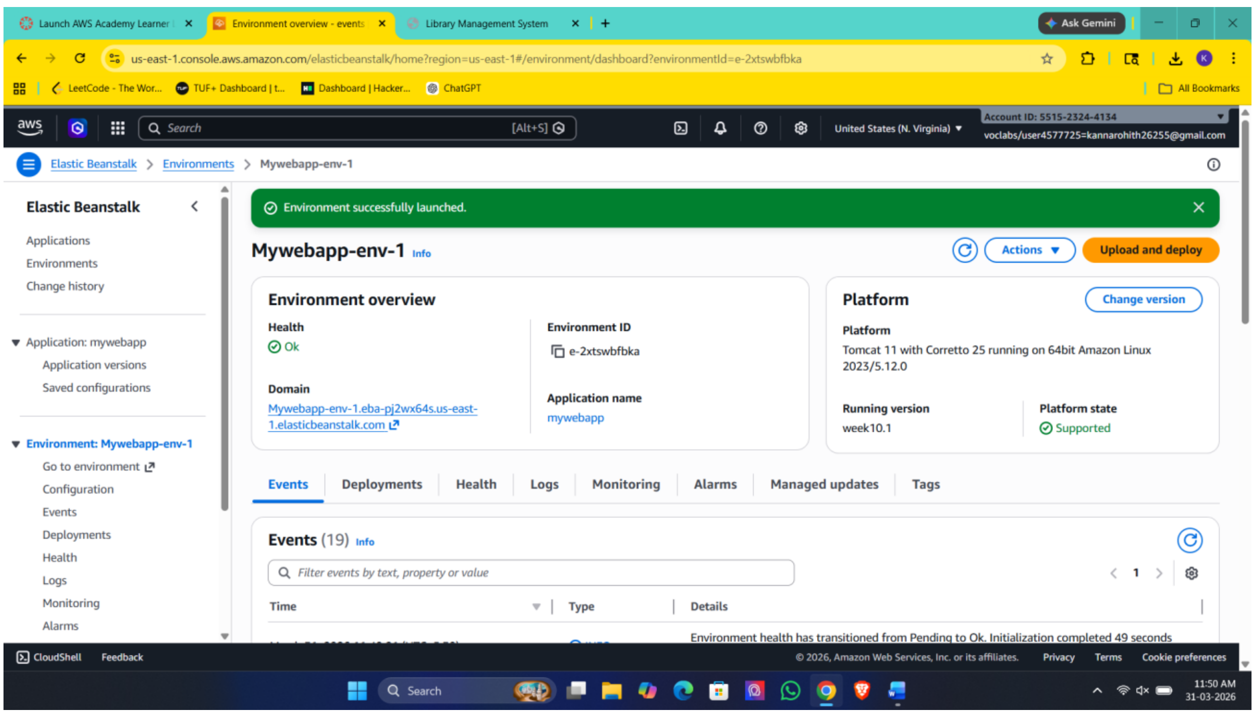
Deployment Process

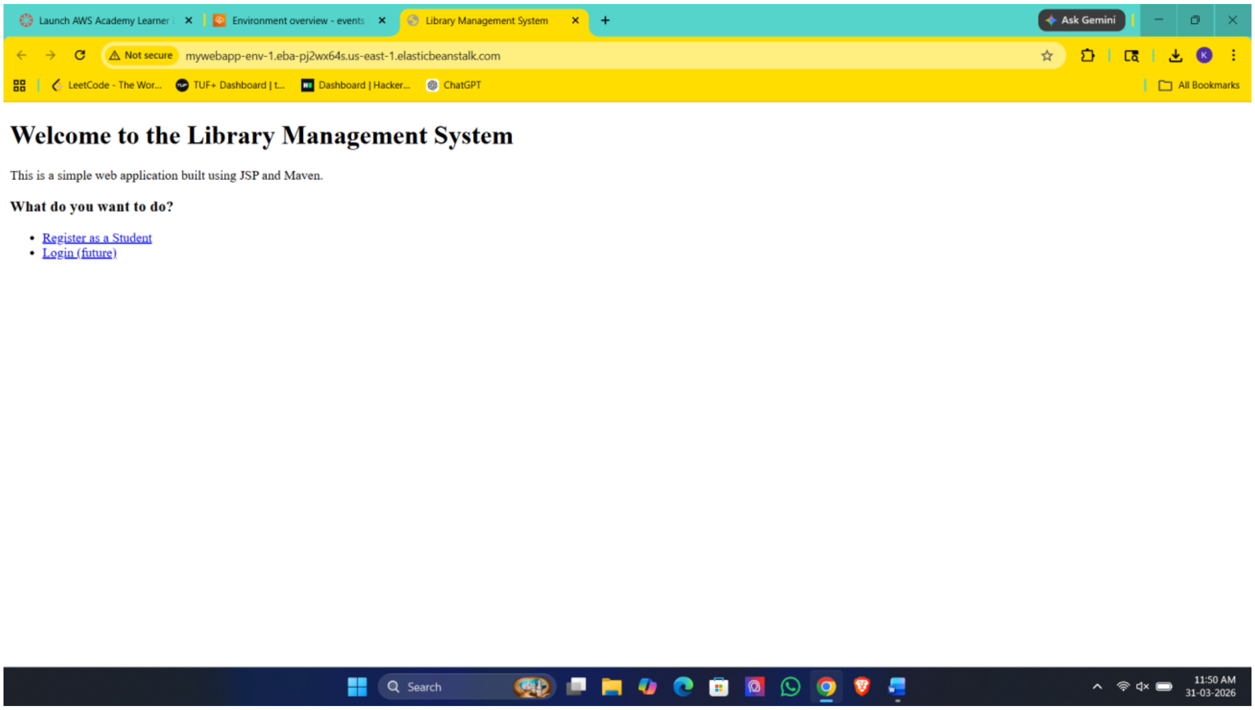
- AWS automatically creates:
    - EC2 instances

- Load Balancer
- Auto Scaling Group
- Security Groups

Access Your Application

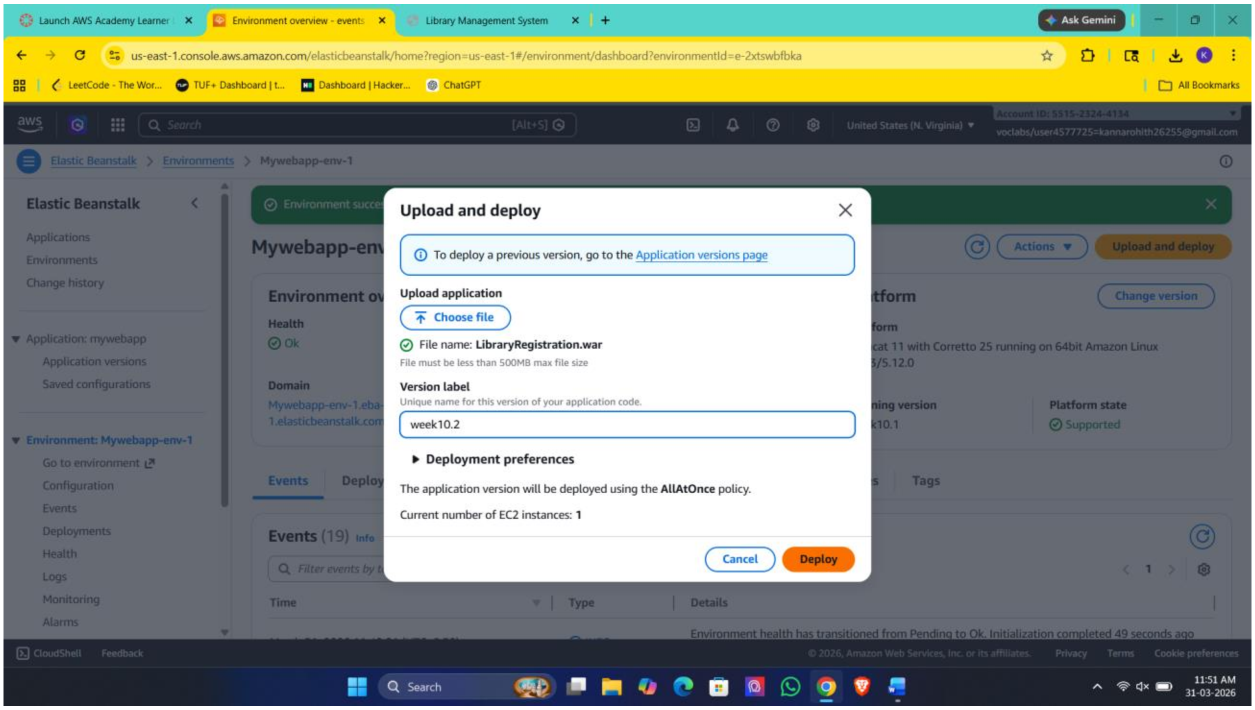
- After deployment:
  - You will get a **URL (domain link)**
  - Open it in browser → Your app will run

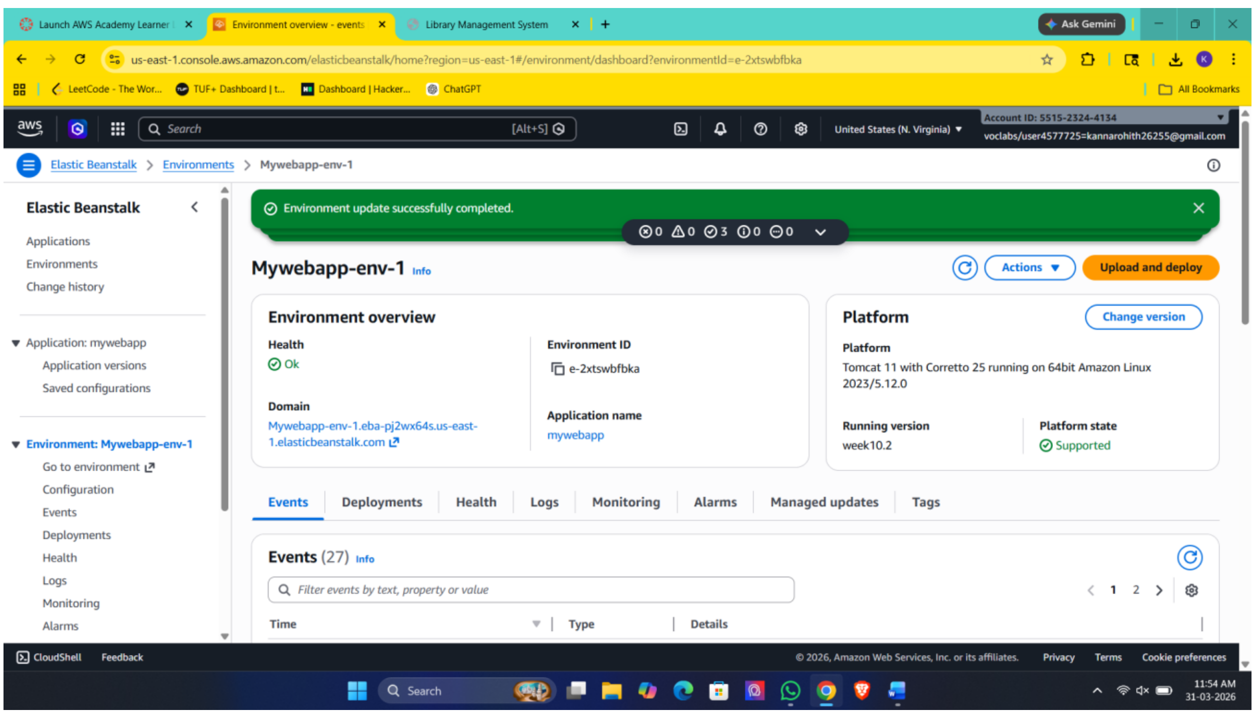


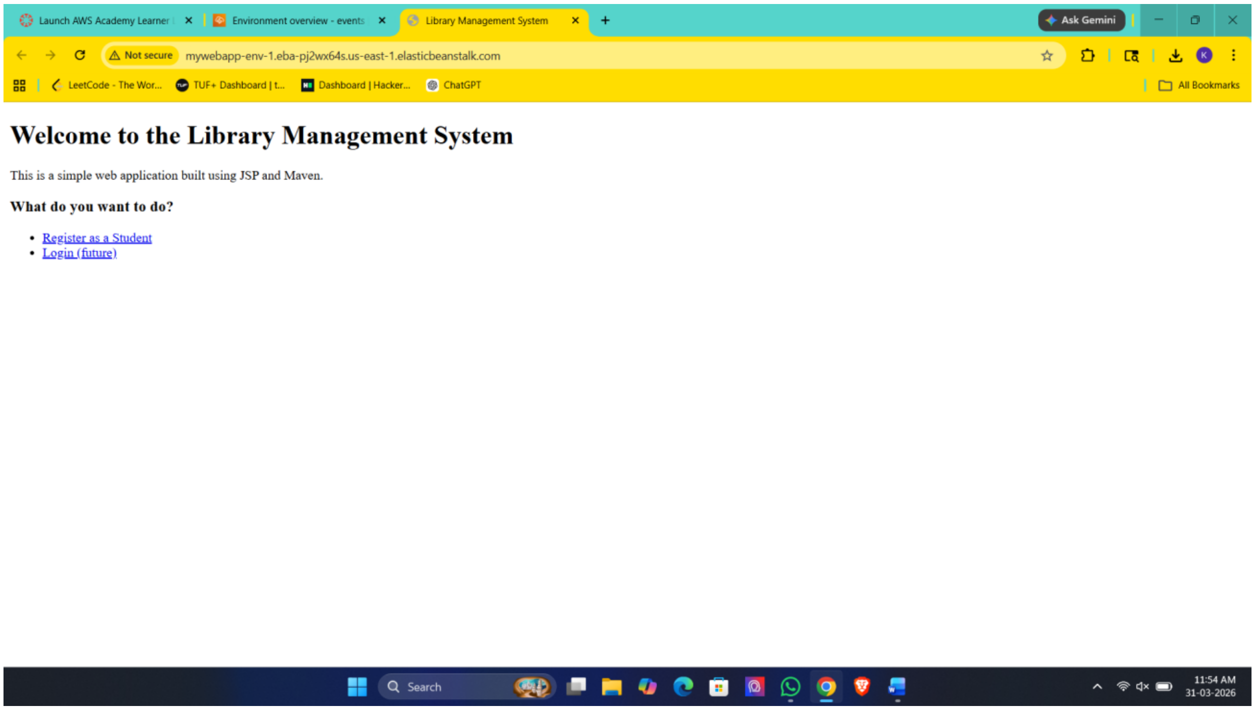


Updating Application

1. Go to your environment
2. Click **Upload and Deploy**
3. Upload new version


Upload and deploy
To deploy a previous version, go to the Application versions page
Upload application
Choose file
File name: LibraryRegistration.war
File must be less than 500MB max file size
Version label
Unique name for this version of your application code.
week10.2
Deployment preferences
The application version will be deployed using the AllAtOnce policy.
Current number of EC2 instances: 1
Cancel
Deploy


Elastic Beanstalk
Environment update successfully completed.
Mywebapp-env-1
Actions
Upload and deploy
Environment overview
Health
Ok
Environment ID
e-2xtswbfbka
Domain
Mywebapp-env-1.eba-pj2wx64s.us-east-1.elasticbeanstalk.com
Application name
mywebapp
Platform
Change version
Tomcat 11 with Corretto 25 running on 64bit Amazon Linux 2023/5.12.0
Running version
week10.2
Platform state
Supported
Events
Deployments
Health
Logs
Monitoring
Alarms
Managed updates
Tags
Events (27)
Filter events by text, property or value
Time
Type
Details



Scenario-Based Questions

Scenario 1: Application Not Loading

Q1. What could be the possible reasons?

- Application crashed or not started
- Wrong port or missing entry file

Q2. Could deployment or configuration be incorrect?
Yes. Wrong platform or incorrect packaging can prevent the app from running.

Q3. How would you troubleshoot this problem?
Check **Events** for failure messages and logs like eb-engine.log to identify startup errors.

Scenario 2: Incorrect Application Structure

Q1. How does application structure affect deployment?
Elastic Beanstalk requires files at the **root level**; incorrect structure prevents detection of the app.

Q2. What happens if files are inside an extra folder?
EB cannot find the entry file, causing deployment failure or blank output.

Q3. How would you fix the structure issue?
Move required files (e.g., .war, app.js) to the root and repackage the ZIP correctly.

Scenario 3: Wrong Platform Selection

Q1. Why is platform selection important?
Platform defines runtime (Java, Node, Python) and how the app is executed.

Q2. What errors might occur?
App won't start or shows runtime errors due to mismatch.

Q3. How can you correct the platform?
Recreate the environment with the correct platform (e.g., Tomcat for .war).

Scenario 4: Environment Health Warning

Q1. Possible reasons for warning?

- Failed health checks
- High response time or errors

Q2. Which sections to check first?
Events, Logs, and Metrics.

Q3. How to resolve?
Fix health check URL, reduce errors, and optimize performance.

Scenario 5: Application Errors in Logs

Q1. What do such errors indicate?
The application is failing during execution.

Q2. Could misconfiguration cause this?
Yes, missing environment variables or resources can break the app.

Q3. How to analyze logs?
Identify the first error, read stack trace, and trace the failing component.

Scenario 6: Deployment Failure

Q1. What could be wrong with the package?
Missing entry file or incorrect ZIP structure.

Q2. Which logs to check?
eb-engine.log and Events tab.

Q3. How do packaging errors affect deployment?
EB cannot start the app, leading to complete failure.

Scenario 7: Works Locally but Not on Cloud

Q1. What environment differences cause this?
OS differences, ports, or file paths.

Q2. Could dependencies/config be responsible?
Yes, missing dependencies or incorrect configs.

Q3. How to debug?
Compare local vs cloud setup and analyze logs.

Scenario 8: Blank Output

Q1. What causes no output?
Routes not defined or frontend not loading.

Q2. Could runtime/config issues be responsible?
Yes, incorrect responses or missing files.

Q3. How do logs help?
Logs show request handling and response errors.

Scenario 9: Application URL Not Accessible

Q1. Could environment still be launching?
Yes, during initialization phase.

Q2. What status indicators to check?
Environment status and health (Pending, Green).

Q3. How can Events help?
They show deployment progress and failures.

Scenario 10: Resource Management Issue

Q1. Cost implications?
Charges continue for running resources.

Q2. Which services run in background?
EC2, Load Balancer, Auto Scaling.

Q3. How to terminate resources?
Delete environment and verify all services are stopped.

Scenario 11: High Traffic Handling

Q1. How does EB handle scaling?
Using Auto Scaling Groups.

Q2. What configurations are needed?
Scaling policies and load balancer.

Q3. What if scaling is not configured?
App slows down or crashes under load.

Scenario 12: Application Update Not Reflected

Q1. Could deployment issues be the reason?
Yes, old version may still be active.

Q2. How can caching affect output?
Browser or CDN may serve old content.

Q3. How to ensure latest version?
Redeploy properly and clear cache.